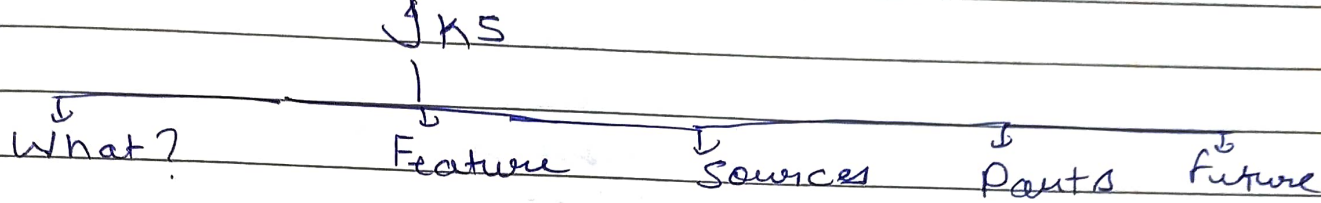
12/8/26 Marvels of Science & Wonder:

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- Foundation of Ancient Indian Times

- Sources of Indian Knowledge System:



10 marks

(i) What is IKS:

- IKS refers to indegenous frameworks of knowledge, developed in ancient India, encompassing a wide range of fields, including philosophies mathematics, linguistics, medicine, astronomy, agriculture, ~~artit~~ architecture, music & governance.

IKS = Indegenous + Multidisciplinary Knowledge (holistic +)

(ii) Feature:

Rooted in holistic and integrated world views. Emphasised experiencial knowledge, ; Intuition, Logical reasing.

- Strong Oral ~~records~~. & Written Traditions (Memory based &

- Guided by Dharma (Moral Order) & Rrtta (Cosmic Order)

↓

ऋत सिद्ध

(रेवाज

(iii) Sources:
- Shruti → Vedas
  - Yajur
  - Rig
  - Atharva
  - Saam
- Smriti → सूत्र, शास्त्र, पुराण, उपनिषद
- ↓
- Objectives
- ↓
- Purushanthas
  - Dharma
  - ~~Arthatha~~ Artha
  - Kama
  - Moksh

Ashram:
- Brahmcharya
- ~~Sgihastha~~ Grahstha
- Vanprasth
- Sanyas

- Vedas → Derived from word Vid which means **to Know**, they are the old Scriptures Can be divided into 4 p
  - Rig Veda → hyms on Natural elem Cosmo
  - Saam Ved → Musical Chanting, basis of Indi Mu
  - Yajurva Ved → Related with rituals & cere
  - Atharva Ved → Related with healing, medicine, practical knowledge.

Scientific Elements in Vedas:
(i) In astronomy: Talks about nakshatra system, motion of stars, mathematics & counting system & large number → Shat Saltpath brahman.

(ii) Medicine: Atharva Ved talked about

(History Timeline)

Date
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healing chant.

(iii) Ecology: Harmony with nature, respect for ~~traverse~~ rivers, forest, acoustics. eg: Sound vibrations of Om, basics of neuroscience.

Shashtras: Systemic text on science & art

Shashtras = To instruct or disciple
- Evolved over centuries during post vedic period.

- Written in prose or verse.
- Covering applied sciences.
eg: Ayurveda → Charak & Sushruth
Arthashashtra
↳ Based on politics & economics
↳ Kautilya

Dhanurveda → Archery
↳ Military science
↳ developed by Vishwa mitra

Nyayashastra → Judiciary

Natyashastra → performing arts & grammar
↳ Bharat muni

Vaastu Shashtra →
Jyotish Shashtra → Astronomy & Astrology
↳ Varha Mihir

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Characteristics of Shashtra :

- → Emphirical Research
- → Interlinking of Philosophy & Practice
- → holistic approach → Body, mind, soul balance

19/8/25 Marvels of Science & Wonder:

IKS: Gyan + Vigyan + Jivana Darshana

Shruti Literature

Vedas

Rig | Saam | Yajur | Atharva

Karm Kand | Gyan Kand

Samhitas | Brahamanas | Aryanakas | Upanishads

Poetic	Prose	Simplified	Vedanta
Mantras	Ritual	Rituals	Discussion on Brahma
Hymns	Sacrificial	Secret Rituals	Jivatma
Chanting	Rites	Scientific	Paramatma
	Explanations	Rite	Unity

विद्या ददाती विनयम् ।

आयुषी मंत्र → Rising Sun god

Smruti (स्मृति)

Vedangas Sutras Dharmashastras Itihasa Pura

Flow of Knowledge:

Shruti → Smriti → Vedanga & Sutra → Purana, Dharmashastra

Commenteries, Debate, Traditions → Culture, Society, Ethics, Science, Spirituality

Timeline:

History
- Pre-History → A ✓, L ✗
- Proto-History → A ✓, L -?
- Historical
  - Ancient → L ✓, A ✓
  - Medieval
  - Modern

Source
- Archaeological
- Literature

Stone Age — 5 Lakh year ago

Indus Valley Civilisation (Sindhu-Saraswati)

Vedic Age
- Early Vedic → Rig Veda
- Later Vedic

6th Century BC → Rise of Buddhism & Jainism

Later Gupta → Harshavardhana

Gupta ← (4 S R K) ← Greek, Shakas, Persian

Post Maurya

Mauryan Empire

a+b

7/09/26

OE Marvels of Science & Wonder:

Sutras = Formula

Veda → Shashtras → Sutras
V → S → S

Foundation → System → Summary

A A NJV
Ayurveda Arthashashtra Natya Shashtra Jyotisha Vastu

Q. Discuss how Vedas, Shastras & Sutras collectively fo
the foundation of ancient Indian Scientific & phile
thought. Provide examples from different disciplin

2/9/26

Marvels of Science & Wonder:

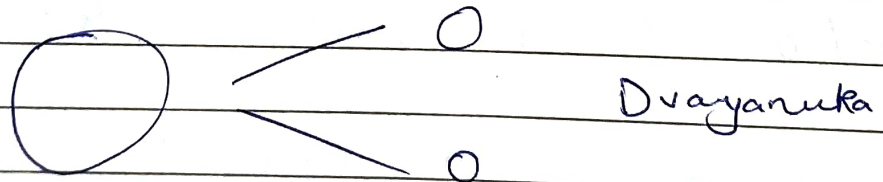
Kanada's Atomism:

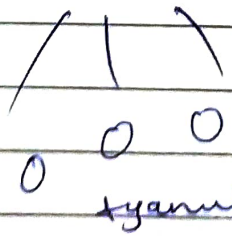
- Kanada - Vaisesika Sutras : (Formula of Specialty)
  - ↳ Kanada
  - ↳ Paramanu
  - ↳ Padarthas
  - ↳ Cosmology

- Kanada was the founder of Vaisesika school he had written a foundation text Vaisesika Sutra
- It talks about earliest systematic atomic theo

Core Ideas of This Theory:

- Universe is composed of indivisible particles (Anu / paramanu)
- Atoms combine to form ~~dvaya Anu~~ dvayanuka & tyanuka





- Categories of Padarth:
  - Dravya (Substance)

Kanada School → Vaisheka
Smallest Particle → Paramanu

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- Guna (Quality)
- Karma (Action)
- Samanya (Generality)
- Visesa (Particularity)
- Samavaya (Inherence)
- Abhava (Non-existent, added later)

- Paramanu: According to Kanada Paramanu is an indivisible, partless, imperceptible material atom.

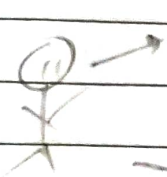
- Five Types of Material Atoms (पंच महाभूत):
  - Prithvi (Earth)
  - Apas (Water)
  - Tejas (Fire)
  - Vayu (Air)
  - Akasa (Ether) → Non-Atomic Element

Vaisesika / Greek philosophy → Parallel

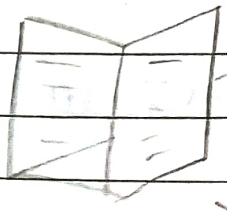
Kanada
↓
Vaisesika Sutra
↓
Padarthas
↓
Paramanu
↓
Atomic Combinations
↓
Composite world
↓
Knowledge of Reality → Moksh

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Brain dump of Kanada:

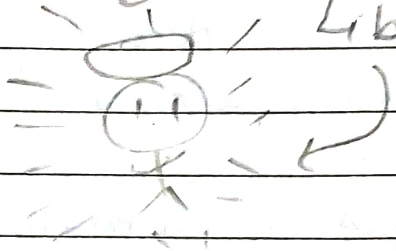


had a book

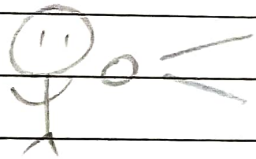


→ Vaisesika

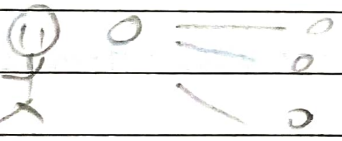
how to get to Moksha/Liberation.



Gave Philosophy about Atom (Anu/Paramanu)



dvayanu



tyiyanu

Categories of Padarth:

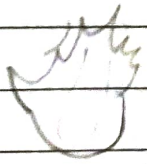
1. 
Substance

Guna — Karma

Sambhanna — Samanya

Abhava — Visesa

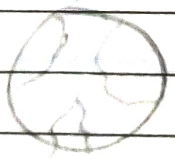
4 material Atoms → Atomic elements



Fire (Tejas)



Water (Apas)



Earth (Prithvi)

Air (Vayu)

Aether (Akasa) → Non-Atomic elements

Daniel Goalman

Stories → Buddha
↳ Yaksha

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Marvels of Science & Wonder:

- Patanjali's Logic & Philosophy:

↳ Yoga Sutras

↳ Padas → Samadhi, Sadhana, Vibhuti, Kaivalya
↓ ↓ ↓ ↓
Not attached Control Liberation

Definition of Yoga:

↳ Clarity & Mastery of mental activity rather than physical activity.

"Yogas citta - Vrittī - Nirodha"
↓ ↓ ↳ Stopping/Restraint
Unite Consciousness Mental Modifications

★ Q. Shruti & Smriti Literature / Sutras & Shasheras thas — 10
Q. IKS & Sources of IKS
★ Q. Patanjali Logic — 5
Q. Kanada Atomic/ Panini Grammar — 5
Q. Historiography of Indian Sciences — 5
★ Q. Overview of Indian Mathematics — 5
↳ Aryabhata concept of Zero

Q. Definition of Yoga → Chitta Vitta Nirodh
Q How many steps in Ashtang → 8

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Patanjali's Logic & Philosophy:

- 2nd Century BCE, Patanjali was author of book Yoga Sutras.
  ↳ had 195 Shlokas.

- Yoga Sutra defines yoga "Chitta Vritti nirodha" (Control of mental modifications.)

- 8 fold path (Astangayoga) → Define the gist of yogic practices.

* Logic & Philosophy:

- It integrated body, mind, spirit as one.
- Psychology of concentration, meditation → define v cognitive Science.
  early

** Nyaya philosophy's link to Patanjali
  ↳ Both shared concern for Logic & Inference & reasoning.

* Modern relevance of Patanjali:
  ↳
  Mindfulness & Meditation in neuroscience
  ↳ Yoga is accepted as a global wellness system through UN Yoga Day.

Michael Denevor

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- Historiography of Indian Science.

- Meaning of Historiography: The study of How history is written & interpreted

- Sources of Indian Sciences:
  - Textual Sources → Vedas → 4 Vedas
    - ↳ Sutras → Yogasutras etc
    - ↳ Shashtras → Rasayan Shashtras
    - ↳ Puranas
    - ↳ Commentaries (Tika) → Commenting something on else's book.
  - Archaeological Sources → Harrappan Seals.
    - ↳ Iron Pillar (Delhi)
    - ↳ Megaliths
      - Big
      - ↳ Stone
        - → Inscription (copper plated)
    - ↳ Epigraphical & Numismatics
      - ↳ Coins

- Historians of Indian Science:
  - Al Biruni
    - ↳ Did translations
    - ↳ Book: Kitab-al-Hind
    - ↳ Written During 11th Century
    - ↳ Describe Indian Astronomy & Mathematics

- Dharmpal was an author → wrote widely on

Thesis + Anti Thesis → gives Synthesis

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Swadeshi

rediscovery of Indian Indegenous Education & Sciences.

- D.D. Kosambi → He had done marxist analysis (gave synthesis) of scientific Indian traditions.

* Colonial Disruption : → British portrayed indian science as "Mythical" knowledge, ignoring empherical (experimental) contributions.

- Contemporary effects TDL : AAYUSH Full Form

CSIR → Council of Science & IKS Division
UNESCO's recognition to Yoga & Ayurved
TKDL → Tradition Knowledge Digital Library

UNIT -2

Ancient Indian Mathematics and Overview :

* ↳ Vedic Mathematics :
- Sulbhasutra → written around 800 B.C.E. provided geometry for altar (Yagyavedi) construction

- Knowledge of $\sqrt{2}$
↳ Pythagorian triplets
↳ π approximation

* Jain Mathematics :
↳ Infinite number, Logarithmic concepts,
↳ Transmission of Knowledge →

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Indian Numerals → Arab → Europe → becomes Global System

- Relevance of Ancient Indian mathematics basis of modern mathematics, coding, finance, space - sciences.

15/9/26 DE : Marvels of Science & Wonder

- Key contributions :

-Decimal System, Place Value
- Algebra (बीज गणित), arithmetic
- Geometry, Combinatorics

Indian Numerals → Arab → Europe → Global

- Aryabhatiya → 121 Verses & 4 Chapters
  - ↳ Written by Aryabhatta
  - ↳ 499 CE.
  - ↳ Maths, time Calculations, Planetary Models & Astronomy
  - ↳ Earth rotates on its axis which explains day & night
  - ↳ Geocentric with some heliocentric Hints
  - ↳ Eclipses as Moon & Earth's shadow

Arya Siddhanta → Aryabhatta? → doubtful

Aryabhatta → Gave Place values, Place holder

* Aryabhatta & Concept of Zero :

Aryabhatta 476 to 550 CE mathematician & Astronomer

Works → Aryabhatiya
→ Arya Sidhant → Doubt?

- Provided tables of Sines trignometric function
- Also provided Latitude & Longitudes concept
- Legacy of Aryabhatiya are used in Indian Panchang & Jyotish Astronomy

Maths works: - Place Value & 0
- Approximation of π
- Sine functions (Jya)

Astronomical Contributions :
- Helio centric Rings.
- Earth Rotates on its Axis
- Explanation of solar - lunar Eclipses (Systems were shadow based)

(Non Mythological)
- Gave Scientific Outlook

Q. Asses the Impact of Aryabhatt & Zero in Global

* Brahmgupta & Bhaskaracharya :
  - Time (घड़ी) → pea (घटिका यंत्र)
  - निमिष → Blinking of eyes
  - Leelavati Sutra :
  - Algebra & Trignometry :

* Brahmgupta → 598 C.E:
  - Brahmsputhit Siddhanta (Khudse dimaag me aaya) (Brahma Ke diya hua gyaan)

gave
- Formal Use of 0, addition, substraction & Negative Numbers
- Quadratic Equations
- Area of Cyclic quadrilaterals

* Bhaskaracharya II : → 1114 CE :
  - Leelavati
  - Beej Ganit
  - Siddhant Shiromani → technical & Advance trignometry

→ Gave concept of Infinity
→ Solution to Pell's equation ($x^2 - dy^2 = 1$) where d is a integer.

→ Trigonometric functions (Sine & Cosines)
→ Gave Calculus like concepts (Rate of change)

Impacts of Indian Mathematics:

↳ Influenced Islamic & European Mathematics

16/9/26 OE Marvels & Science

~~Tarab's Complement~~

- Surya Siddhanta & Cosmology:

↳ Astronomy, time, Planetary Motion, Cycles of Cosmos

Written by:

Latadeva / Laatdeva (लाटदेव) During 4-5th CE

↳ Student of Aryabhatta -I

Surya Sidhant

↳ Planetary Motion & Computation of Celestial Positions

↳ Time calculations, Trignometry

Equinoxes - Equal day & Night

Solstices - Either day or Night are long

↳ Explains solar & Lunar Cycles

* Astronomical Insights:

- Calculation of Earth's diameter; App. 12742 Km Very close to modern calculation

- Calculated solar year (365.256 days)

* Philosophical:

↳ Talks about; that universe is Cyclic in time (Kala Chakra)

* Influence of Surya Siddhant:

↳ Based of Panchanga

↳ Principles of Navigation & travel